

ServicePlus 2.0 Analytics & Forecasting

Internship Project Presentation

Submitted by

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8 June – 31 July 2026

Transforming ServicePlus transactional data into analytical insights, operational dashboards and future-demand forecasts.

ServicePlus 2.0: Brief

What is ServicePlus 2.0?

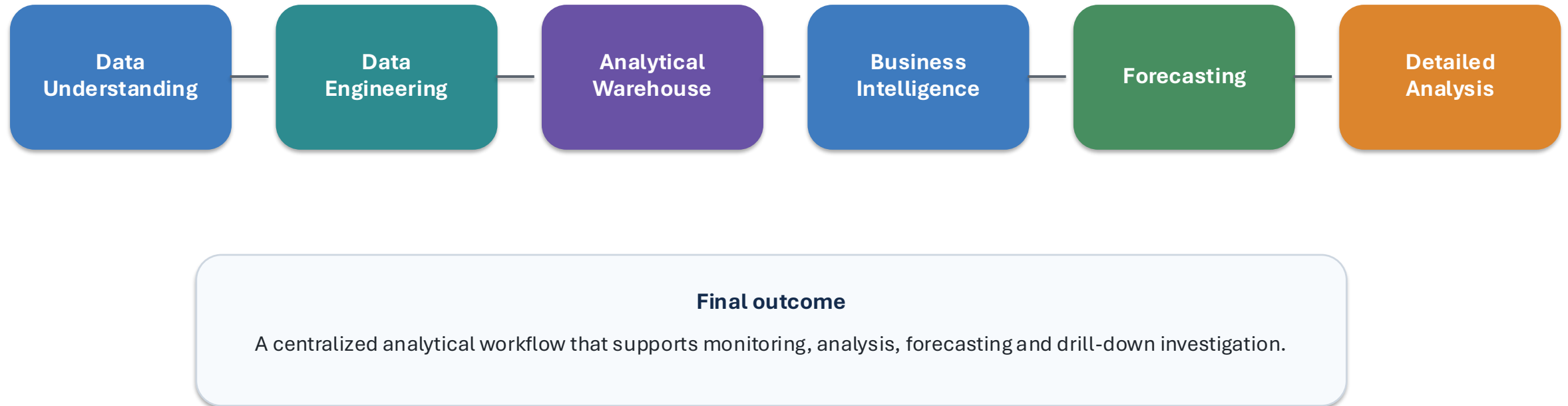
ServicePlus 2.0 is NIC's e-Governance service delivery platform used by government departments to design, deploy and monitor online citizen services.

The platform generates operational data across:

- Citizen applications and service requests
- Service and department information
- Application status and workflow stages
- Processing and completion times
- SLA-related information
- Applicant and geographical information

What is the Project About?

From Raw Service Data to Decision-Ready Analytics



Phase 1: Data Discovery & Profiling

Understanding the Source Data

Before building dashboards or models, I first established what information was available and how it could be used for analytics.

What I worked on

- Reviewed application and service information.
- Identified department and workflow-related information.
- Examined applicant and geographical fields.
- Identified application status and processing information.
- Reviewed submission and completion dates.
- Mapped the relationships required for later analysis.

Objective

Establish a clear understanding of the available data before beginning transformation and dashboard development.

Foundation for later phases

The profiling work informed the warehouse design, ETL rules and downstream analysis.

SOURCE DATA OVERVIEW – What Was Present in the Table

Table: data_stats | Total Records: 2,132,046

1. SERVICE INVENTORY



Total Applications

2,132,046



Total Unique Services (base_service_id)

35



Service Type Identified

- Aadhaar Verification Services (State-wise)



Each record represents a service application processed through ServicePlus 2.0.



Key information is stored inside JSON fields in the table.



The data captures the complete lifecycle of an application – from initiation to execution and final outcome.

2. STATUS INVENTORY

I

I : INITIATED / IN PROGRESS

F

F : FINALIZED

D

D : DISPOSED / DELIVERED

R

R : REJECTED

3. initiated_data JSON STRUCTURE

initiated_data JSON contains 3 major sections:



appl_info

Application and service related information



next_task_data

Information about the next stage in the workflow



application_form_attributes

Information submitted by the applicant in the application form

3.1 appl_info – Key Fields

- appl_id
- appl_ref_no
- service_id
- service_name
- base_service_id
- department_id
- department_name
- user_id
- user_name
- submission_date
- submission_mode
- submission_location
- submission_location_id
- submission_location_type
- due_date

3.2 next_task_data – Key Fields

- next_task_id
- next_task_name
- next_task_type
- next_user_id
- user_name
- user_login
- designation
- department_level
- location_id
- location_name
- location_type_id

3.3 application_form_attributes – Key Fields

- Applicant Name
- Address
- PIN Code
- Date of Birth
- Image Path
- Village
- (Other form-specific fields)



PII Identified

- Applicant Name
- Address
- Photograph Path

4. execution_data JSON STRUCTURE

execution_data JSON contains 3 major sections:



task_info

Details of the current task being executed (action, time, officer etc.)



next_task_data

Information about the next workflow stage after the current task



official_form_attributes

Information filled by the official during task processing

4.1 task_info – Key Fields

- task_id
- task_name
- task_type
- action_no
- action_taken
- action_detail
- received_time
- executed_time
- user_name
- designation
- location_id
- location_name
- department_level
- next_task_id
- next_task_name

4.2 next_task_data – Key Fields

- next_task_id
- next_task_name
- next_task_type
- next_user_id
- user_name
- user_login
- designation
- location_id
- location_name
- department_level
- location_type_id

4.3 official_form_attributes – Key Fields

- Official form related attributes captured during task execution (varies by service)



Key Takeaway

The source data provides both application-side information (who, what, where, when) and workflow-side information (how it was processed, by whom, and what happened next).



Purpose of Extraction

Extract these JSON fields into structured columns to enable analytics, dashboards, SLA monitoring and demand forecasting.

Phase 1: Data Quality & Profiling

Checking Whether the Data Was Ready for Analysis

Completeness

- Null values and null ratios
- Missing dates
- Temporal completeness

Consistency

- Unexpected values
- Inconsistent structures
- Invalid categories

Uniqueness

- Duplicate records
- Cardinality of important fields

Relationships

- Entity relationships
- Key consistency
- Referential dependencies

DATA QUALITY PROFILING – KEY FINDINGS

Profiling performed on ServicePlus dataset (Table: data_stats) with 2,132,046 total records

1. NULL ANALYSIS

Presence of execution_data



■ execution_data present
282,273
(13.24%)

■ execution_data missing
1,849,773
(86.76%)



execution_data is available only after an application enters the execution/workflow stage. Therefore, missing values are expected and not treated as data quality issue.

2. CARDINALITY ANALYSIS



Unique Applications
710,682



Unique Reference Numbers
710,682



Unique Services
35



Unique Statuses
4



Total rows = 2,132,046
One application can have multiple records.

3. SERVICE ANALYSIS



35
Unique Services

Top 5 Services by Applications

Service (Example)	Applications	% Contribution
Aadhaar Verification by State Portal – Maharashtra	91,529	12.95%
Aadhaar Verification by State Portal – Andhra Pradesh	67,639	9.58%
Aadhaar Verification by State Portal – Karnataka	52,300	7.40%
Aadhaar Verification by State Portal – Kerala	46,847	6.64%
Aadhaar Verification by State Portal – Uttar Pradesh	40,481	5.72%

Highest Contribution
Maharashtra
91,529
(12.95%)

Lowest Contribution
Lakshadweep
5
(0.00%)



Significant variation in service-wise demand across states/service variants.

4. OUTLIER ANALYSIS

Service-level Outliers

Highest
Maharashtra
91,529

Lowest
Lakshadweep
5

Volume Outliers (by month)

Month	Applications
Highest Volume Month	292,534
2nd Highest	216,495
3rd Highest	136,997
Lowest Volume Month	14,396
2nd Lowest	13,196



Application volumes vary significantly across months with major spikes and low activity periods.

KEY TAKEAWAYS



Stage-dependent missingness observed in execution_data (available only post initiation of execution).



710,682 unique applications from 2,132,046 total records indicates multiple records per application.



35 unique services with high variation in demand across states/service variants.



Presence of service-level and monthly volume outliers that need to be considered in analysis and forecasting.



Cardinality analysis shows 4 statuses, ensuring clear classification of application lifecycle stages.

DEPARTMENT ANALYSIS & DATA QUALITY ISSUES

Analysis performed on ServicePlus dataset (Table: data_stats) | Total Records: 2,132,046

DEPARTMENT ANALYSIS



Department-wise application distribution based on [unique applications](#)

Rank	Department Name	Unique Applications	% Share
1	Revenue Department	319,487	44.85%
2	UIDAI State Verification	281,187	39.49%
3	Revenue and Disaster Management Department	39,455	5.56%
4	Department of Revenue, Rehabilitation and Disaster Management	31,408	4.41%
5	UIDAI	24,721	3.47%
6	DEPARTMENT OF REVENUE	14,424	2.02%

Top 3 Departments (by unique applications)



1. Revenue Department

319,487
(44.85%)



2. UIDAI State Verification

281,187
(39.49%)



3. Revenue and Disaster Management Department

39,455
(5.56%)



Top 2 departments (Revenue Department + UIDAI State Verification) contribute **84.34%** of total unique applications.



KEY TAKEAWAY

Department-wise analysis shows high concentration in top departments. However, inconsistent naming causes duplication and fragmentation. Standardization is essential for reliable analytics.

DATA QUALITY ISSUE IDENTIFIED



Inconsistent Department Naming

Same or similar departments appear with different names, causing the same department to be counted multiple times.

Examples:

- Revenue Department
- DEPARTMENT OF REVENUE
- UIDAI
- UIDAI State Verification

→ This leads to fragmentation and inflated counts for departments.



Action Required

Standardize department names and map them to a master list for accurate aggregation and reporting.

Phase 2: Architecture & Process Workflow

Designing the End-to-End Analytics Flow

The architecture defines how the data moves from the ServicePlus source to the final analytical outputs.

Architecture

ServicePlus Source

↓

Raw / Staging Data

↓

Profiling & Processing

↓

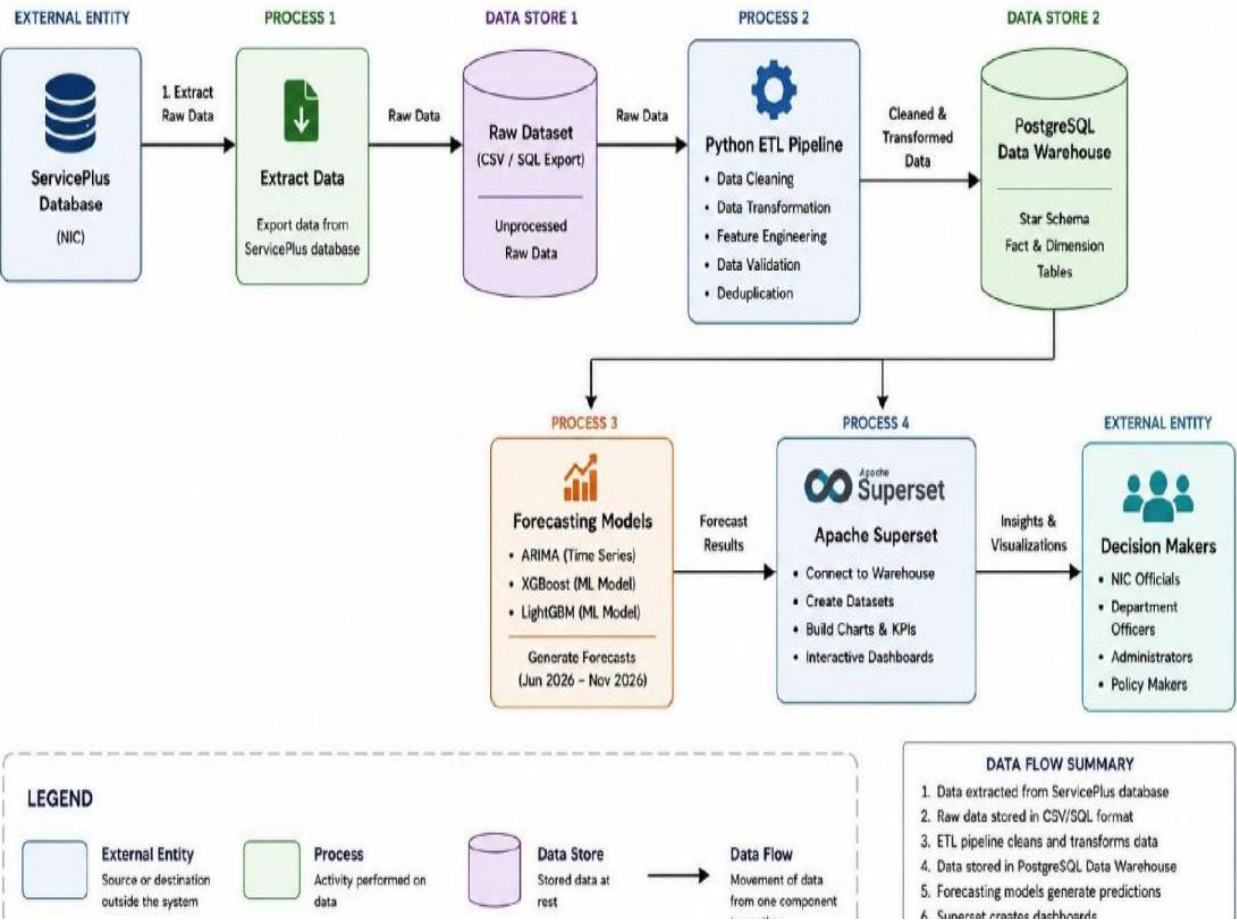
Python ETL

↓

Quality Gate

↓

PostgreSQL Analytical Warehouse



The warehouse feeds both BI dashboards and forecasting, with Streamlit providing detailed analysis.

Phase 3: Data Flow

¹² Extraction

ServicePlus transactional data is brought into the analytics workflow.

¹² Staging

Raw data is kept separately before transformation.

¹² Transformation

Python-based processing performs cleaning, transformation, validation and deduplication.

¹² Loading

Validated data is loaded into the PostgreSQL Star Schema.

¹² Analytical Processing

The warehouse feeds dashboards and forecasting.

¹² Consumption

Insights are presented through dashboards and detailed analysis.

Why this structure was important: It separates raw operational data from analytical processing and creates one consistent analytical source for downstream work.

1. ETL – EXTRACTION



Objective:

Extract application and workflow data from ServicePlus 2.0 source for further processing.



SOURCE

- Table: data_stats
- Total Records: ~ 2,132,046



EXTRACTION LOGIC (Python – etl.py)

```
from sqlalchemy import create_engine
import pandas as pd

engine = create_engine(
    'postgresql+psycopg2://user:password@localhost:5432/nic_db',
)

query = """
SELECT
    application_id,
    appl_ref_no,
    base_service_id,
    appl_status,
    submission_date
FROM public.data_stats
LIMIT 10
"""

df = pd.read_sql(query, engine)
print(df.head())
print(df.info())
```

TEST EXTRACTION (10 ROWS)

application_id	appl_ref_no	base_service_id	appl_status	submission_date
32493412	1092101529882222002	1950	F	2025-10-07 14:22:32
32493788	109210133799822120312533	1950	F	2025-10-07 14:29:47
32503876	1092101423615620812082316217	1950	F	2025-10-07 17:39:33
...	...	...	...	...

FIELDS EXTRACTED

Field Name	Description	Data Type
application_id	Unique application identifier	int64
appl_ref_no	Application reference number	str
base_service_id	Base service identifier	int64
appl_status	Application status	str
submission_date	Date & time of submission	datetime64[us]



Total Records in Source: ~ 2.13 million

2. ETL – TRANSFORMATION



Objective:

Clean, standardise and prepare the extracted data for analytical use.

TRANSFORMATION WORKFLOW



Extracted
Data



Clean &
Standardise



Inspect &
Validate



Deduplicate &
Prepare



Transformed
Data

2.1 DEPARTMENT NAME STANDARDISATION

Before Cleaning (6 Unique Values)	After Cleaning (2 Standard Values)
- Revenue Department - Revenue and Disaster Management Department - Department of Revenue, Rehabilitation and Disaster Management - DEPARTMENT OF REVENUE - UIDAI - UIDAI State Verification	- Revenue Department - UIDAI



Departments: 6 → 2

Outcome: Standardised names reduced duplicates and improved consistency across the dataset.

2.2 SERVICE NAME STANDARDISATION (EXAMPLE)

Before Cleaning	After Cleaning
Aadhaar Verification by State Portal – Assam	Aadhaar Verification by State Portal – Assam
Aadhaar Verification by State Portal - Assam	Aadhaar Verification by State Portal – Assam
Aadhaar Verification by State Portal – assam	Aadhaar Verification by State Portal – Assam
Aadhaar Verification by State Portal – Assam	Aadhaar Verification by State Portal – Assam



Outcome: Service names standardised to a single consistent format.

2.3 OTHER TRANSFORMATION ACTIONS



JSON Extraction

Extracted important fields from JSON columns (appl_info, execution_data) using Python.



Data Cleaning

Handled null values, corrected data types, trimmed spaces and removed invalid values.



Data Validation

Validated mandatory fields, date formats, and logical consistency of values.



Deduplication

Removed duplicate records based on unique application identifiers.



Analytical Field Preparation

Prepared fields for aggregation, KPI calculation and forecasting.



Key Takeaway:

Transformed raw, inconsistent data into clean, standardised and analysis-ready data by resolving naming variations, extracting JSON fields, validating data quality and removing duplicates.

3. ETL – LOADING



Objective:

Load the transformed and validated reporting datasets into PostgreSQL and verify the data in DBeaver.



Transformed & Validated Data



PostgreSQL



Verified in DBeaver

REPORTING TABLES LOADED

- report_status_distribution
- report_service_distribution
- report_monthly_trend
- report_workflow_actions
- report_department_distribution

EXAMPLE: report_status_distribution

Load Query (Python – etl.py)

```
df.to_sql(  
    "report_status_distribution",  
    engine,  
    schema='analytics',  
    if_exists='replace',  
    index=False  
)  
print("Table Loaded Successfully")
```

Verification in DBeaver

appl_status	applications
I	1,849,776
F	154,314
D	118,593
R	9,363

* Sample output in DBeaver after loading

ALL REPORTING TABLES (SAMPLES)

report_service_distribution (Top 5)	
service_standardized	applications
Aadhaar Verification by State Portal – Rajasthan	1
Aadhaar Verification by State Portal – Sikkim	1
Aadhaar Verification by State Portal – Tamil Nadu	1
Aadhaar Verification by State Portal – Telangana	1
Aadhaar Verification by State Portal – Tripura	1

* Total unique services: 35

report_monthly_trend (Sample)	
month	applications
2024-01-01	42
2024-02-01	21
2024-03-01	87
...	...

* Trend based on submission_date

report_workflow_actions	
action_detail	occurrences
Forward	182,727
Deliver	93,474
Reject	6,045
NULL	27

* Total workflow actions captured

report_department_distribution	
department_standardized	applications
Revenue Department	1,214,322
UIDAI	917,724

* Top Departments (Standardised)



Load Status: All reporting tables loaded successfully into PostgreSQL.



Validation Status: Row counts verified, data types validated and samples checked in DBeaver.

Phase 4: PostgreSQL Data Warehouse

Designing the Analytical Data Model

The PostgreSQL warehouse was designed to provide a structured and reusable source for:

- Dashboard queries
- Filtering and aggregation
- SLA analysis
- Service-level analysis
- Forecasting features

Central Fact Table

fact_service_transaction

Grain: One record represents one application / service-request transaction.

Connected dimensions

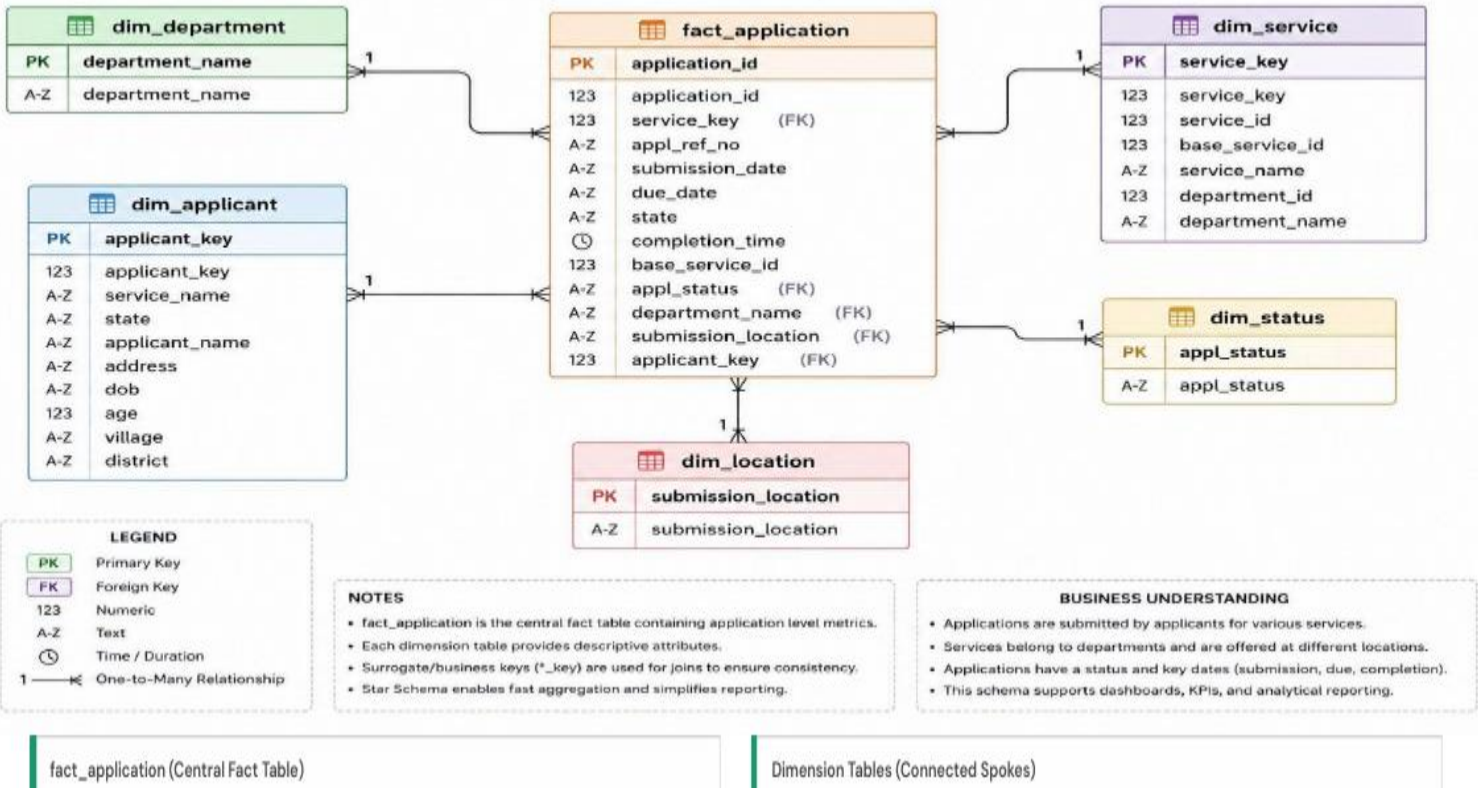
- Applicant
- Service
- Department
- Location
- Status

Design principle

This structure was designed around the natural grain of a citizen service transaction.

Phase 5 : Star Schema

ER DIAGRAM – APPLICATION ANALYTICS DATA WAREHOUSE
(Star Schema)



FACT

fact_service_transaction

- Contains:
- Application ID
 - Service key
 - Applicant key
 - Department key
 - Location key
 - Status key
 - Submission date
 - Completion date
 - Processing hours
 - SLA status

Why I used a Star Schema

It provides a clear analytical structure, reduces complex joins and allows the same warehouse to support both dashboards and forecasting.

DIMENSIONS • dim_service • dim_applicant • dim_location • dim_department • dim_status

Phase 6: Business Intelligence Dashboard

Building the Interactive Analytics Layer

The prepared PostgreSQL warehouse was connected to Apache Superset to create the interactive dashboard.

Overview

Applications

Applicants

Task Processing

Service Performance / SLA

Forecasting & Trends

Common dashboard functionality

- State
- Service filter
- Submission-date filter
- Apply / Clear All
- Drill-down analysis

Purpose: Convert the prepared analytical data into an interactive interface for operational monitoring and decision-making.

Overview —
Understanding the ServicePlus Landscape

35

ACTIVE STATES

35

AVAILABLE SERVICES

711K

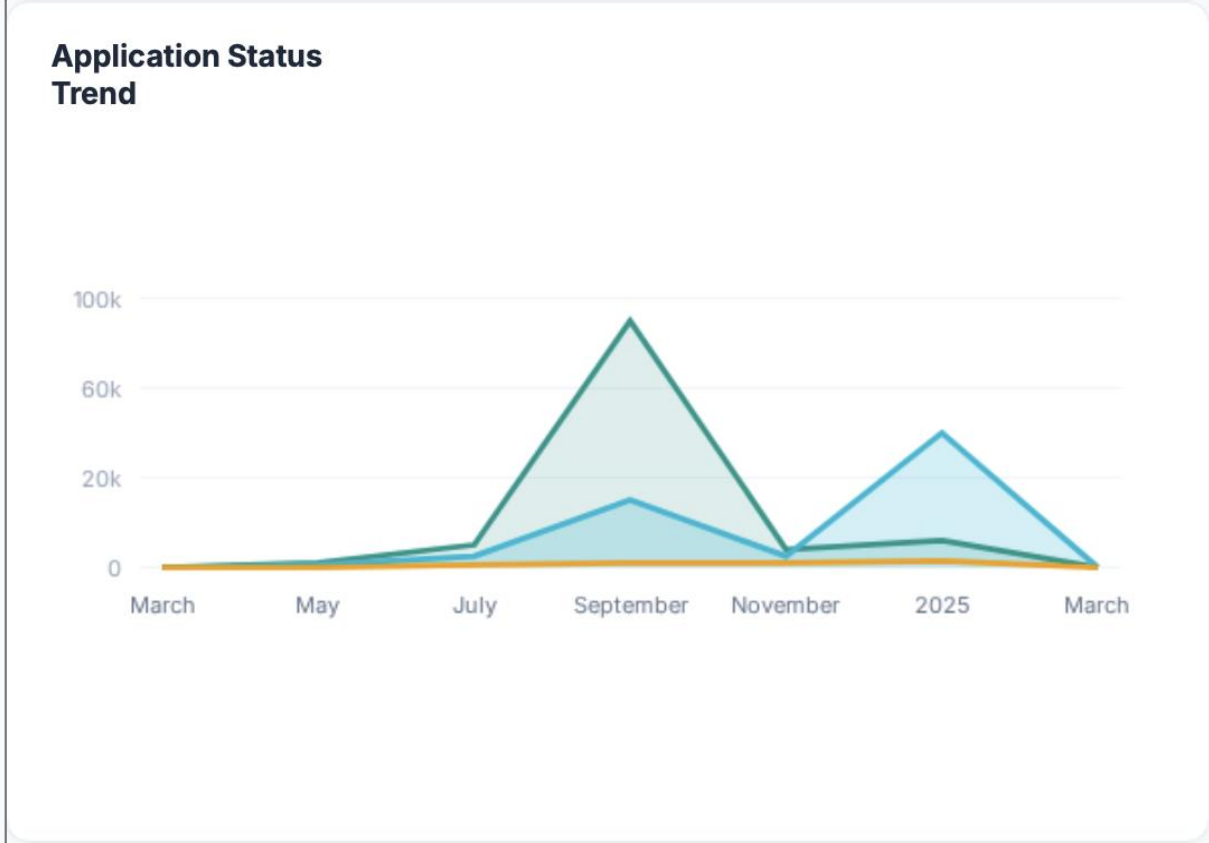
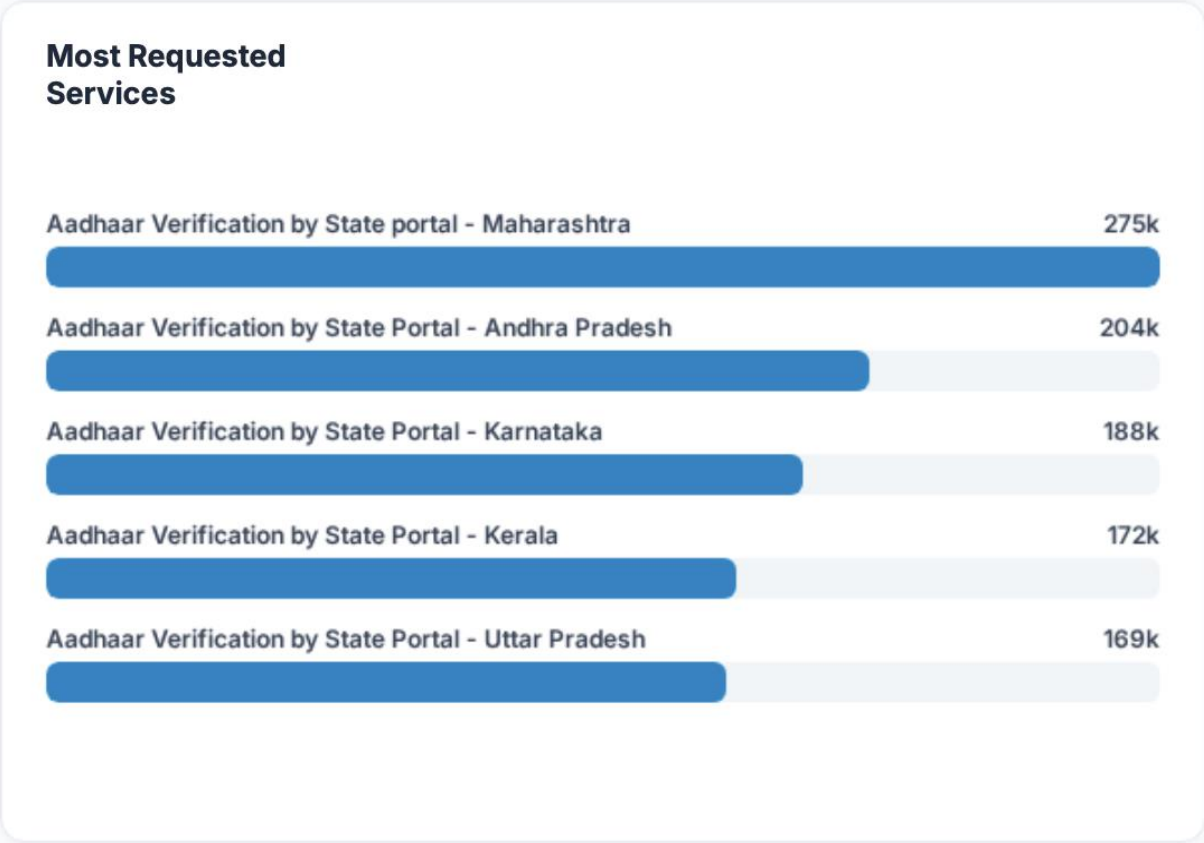
TOTAL APPLICATIONS

668K

IN PROGRESS

42.7K

PROCESSED



01

ESTABLISH THE BASELINE
711K applications across 35 services and 35 active states.

→

02

EXAMINE CURRENT STATUS
668K applications remain in progress, compared with 42.7K processed.

→

03

DEFINE THE NEXT QUESTION
The baseline established where to investigate application demand next.

Applications —

Identifying Demand Patterns

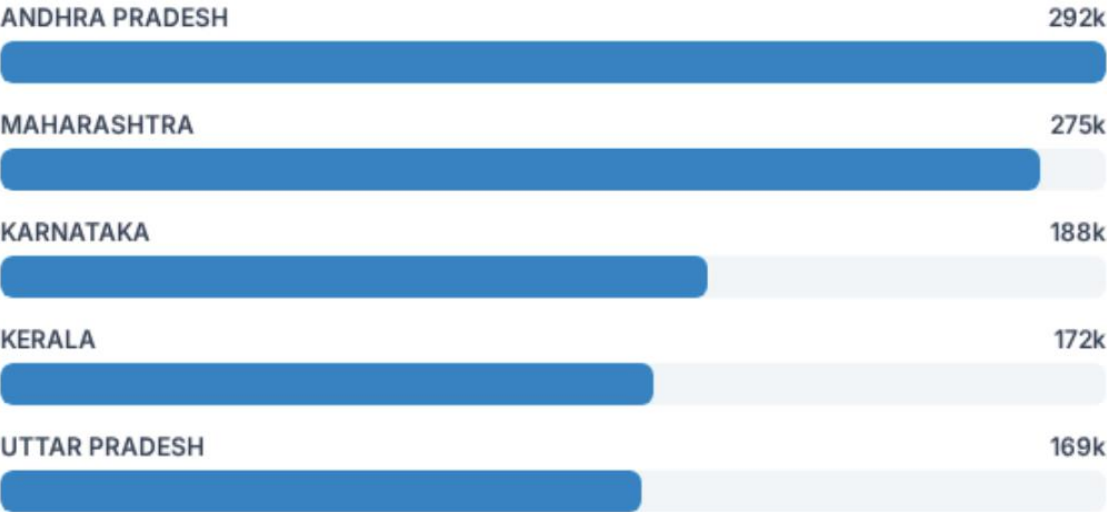
711K
APPLICATIONS

42.7K
PROCESSED

668K
IN PROGRESS

35
SERVICES

Top States by Applications



What the demand pattern says

Andhra Pradesh — **292K**
Karnataka — **188K**
Uttar Pradesh — **169K**
Maharashtra — **275K**
Kerala — **172K**

MAIN INSIGHT
Application demand is concentrated across a small group of high-volume states, shaping downstream workload.

01 BREAK DOWN DEMAND
Application activity was analysed across services, departments, states and time.



02 IDENTIFY VARIATION
Application volumes differed substantially across services and departments, with notable monthly fluctuations.



03 FOLLOW THE WORKLOAD
These demand patterns led to the next question: how did the resulting workload move through the workflow?

Task Processing —

Understanding Operational Workload

533K

TOTAL TASKS

119K

COMPLETED

9.38K

REJECTED

1.27K hrs

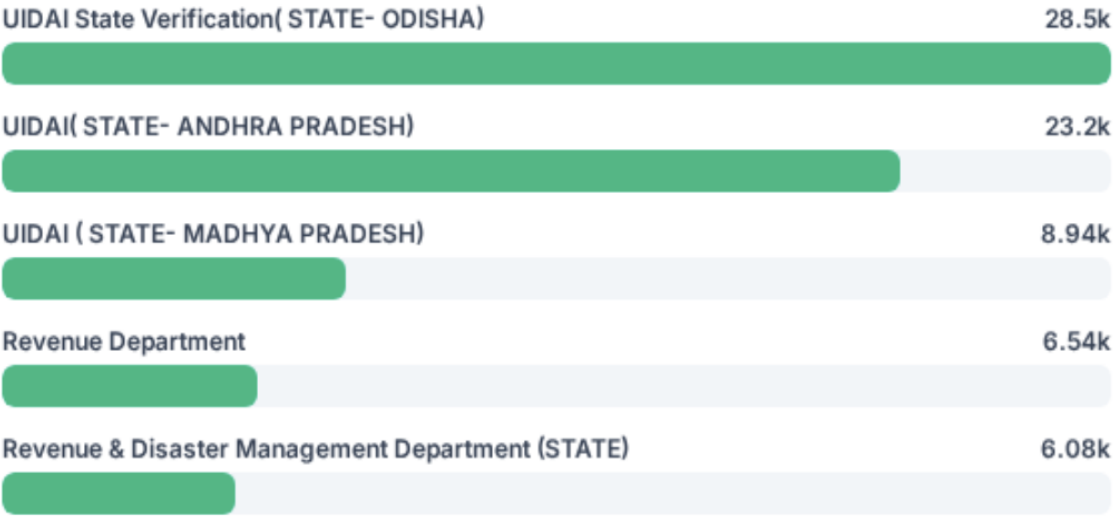
AVG. PROCESSING

Task Outcome Distribution



- Forward
Workflow movement
- Deliver
Final completion
- Reject
Rejected outcomes

Top Verification Offices by Workload



01

TRACE APPLICATION ACTIVITY

533K tasks represented the workflow activity associated with the application workload.

02

EXAMINE OUTCOMES

Forward, Deliver and Reject actions showed how applications progressed through processing.

03

IDENTIFY WORKLOAD DIFFERENCES

Workload and processing time varied across verification offices and organisational levels.

Service Performance —

Measuring Delivery Efficiency

376K

WITHIN TARGET

157K

BREACHED

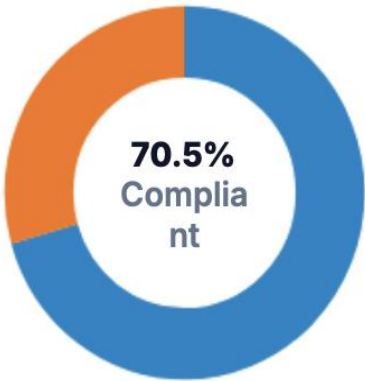
70.48%

SLA COMPLIANCE

1.27K hrs

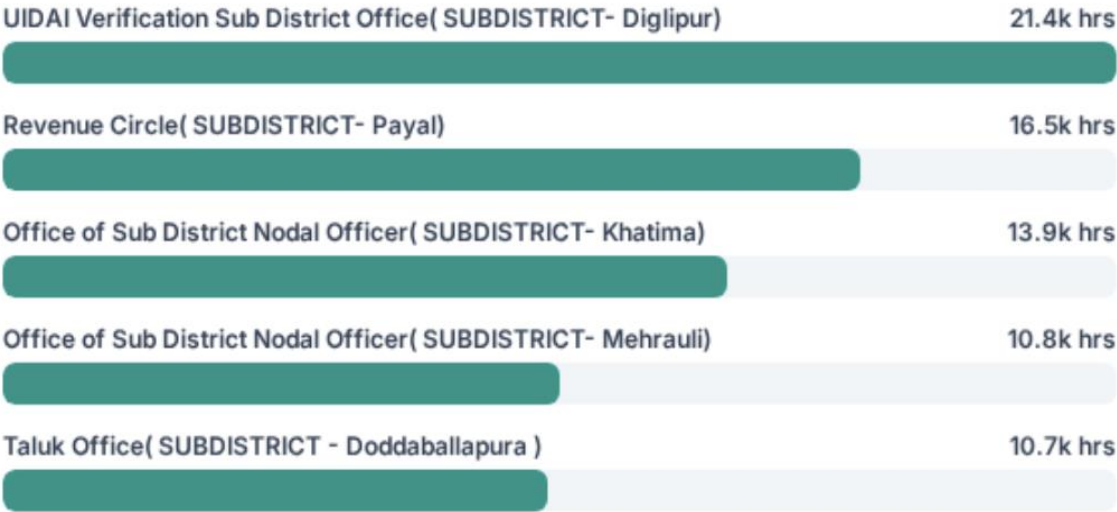
AVG. COMPLETION

SLA Compliance Distribution



- Within SLA: 376k
Target met (70.48%)
- Breached SLA: 157k
Exceeded target (29.52%)

Top Offices by Processing Time



01

MEASURE SLA COMPLIANCE

70.48% were within target time; 376K met the target and 157K exceeded it.

02

MEASURE PROCESSING TIME

Average completion time was approximately 1.27K hours.

03

IDENTIFY EFFICIENCY GAPS

SLA breaches and processing-time differences highlighted where efficiency could improve.

Forecasting & Trends —

Anticipating Future Demand

371K
JUNE FORECAST

352K
JULY FORECAST

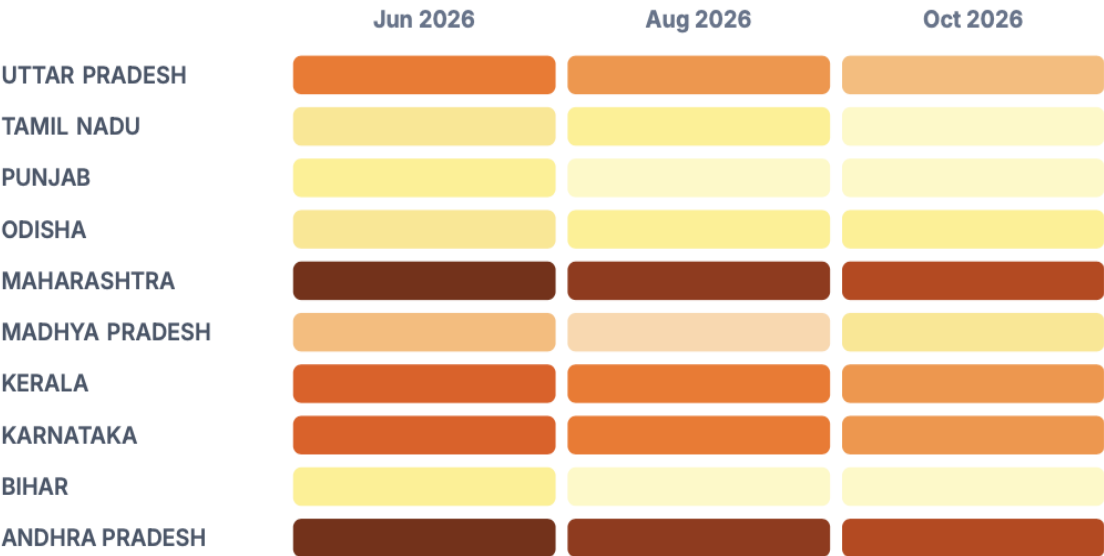
-5.19%
JUNE → JULY

333K
NOVEMBER FORECAST

Application Forecast Trend (Jun–Nov 2026)



Application Demand Heatmap (Jun–Nov 2026)



01 ESTIMATE UPCOMING DEMAND
Forecast indicated approximately 371K applications for June 2026 and 352K for July.

02 IDENTIFY THE DIRECTION
June-to-July demand was expected to decline 5.19%, trending toward 333K by November.

03 LOCATE FUTURE DEMAND
State/service heatmaps showed where expected future application demand was concentrated.

Forecasting: Geographic Demand & Confidence



Application Demand Heatmap

Shows how expected demand varies across states and months.

State-wise Expected Applications

June 2026 forecast identifies higher-demand states, including:

- Andhra Pradesh — 67.4K • Maharashtra — 63.3K • Karnataka — 43.5K
- Kerala — 39.6K • Uttar Pradesh — 38.9K

Forecast Confidence Interval

The forecast is presented with:

- Upper bound
- Forecast
- Lower bound



Variation and uncertainty are communicated alongside the forecast.

Phase 7: Streamlit Integration

From Dashboard KPI to Detailed Analysis

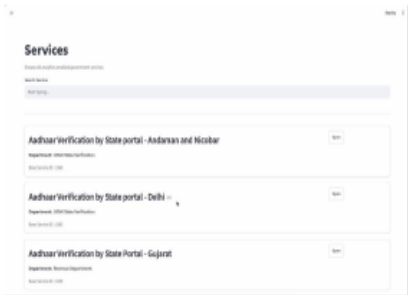
The dashboard provides the high-level view, while Streamlit provides deeper analytical exploration.

 **Service KPI
(Superset)→**



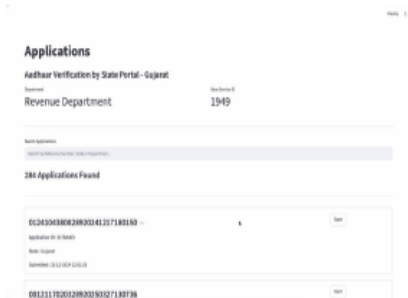
Click service performance KPI

 **Services Page
(Streamlit)→**



Browse service-wise analytics
and interact with filters

 **Application Details
(Streamlit)**



Explore detailed
application-level information
and insights

Workflow

- Superset Dashboard
- ↓
- User selects KPI / visualization
- ↓
- Relevant context is passed through URL parameters
- ↓
- Streamlit analysis page opens
- ↓
- Streamlit queries PostgreSQL
- ↓
- Detailed charts and analysis are displayed

Phase 8: Outcomes & Deliverables

What Was Delivered

D1 — Data Profiling Report

D2 — Star Schema Design

D3 — ETL Pipeline

D4 — SLA Compliance Dashboard

D5 — Service Demand Heatmap

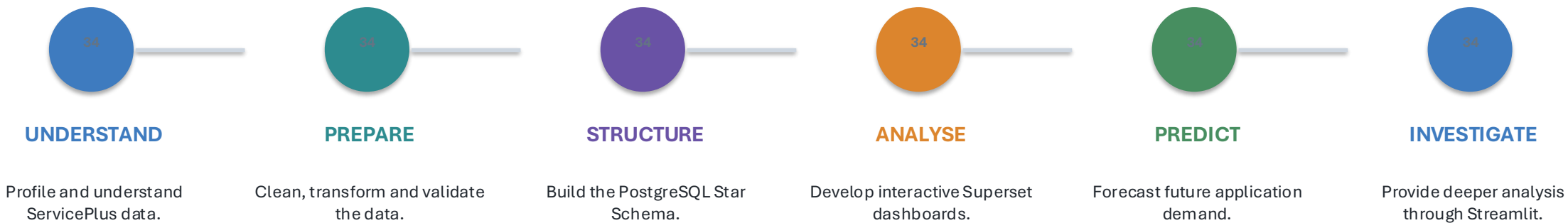
D6 — Forecast Model & Model Card

D7 — Streamlit Integration

D8 — Final Report & Documentation

Final Project Summary

From Raw Data to Decision Support



The project transformed ServicePlus transactional data into a structured analytics and forecasting solution capable of supporting operational monitoring, service-performance analysis and future-demand planning.

Thank You

ServicePlus 2.0 — Analytics & Forecasting

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National Informatics Centre